



Department of Information Technology

Academic Year 2024 – 2025 (Odd Semester)

Degree, Semester & Branch : V Semester-B. Tech Information Technology
Course Code & Title : CS3591 Computer Networks
Name of the Faculty member (s) : Dr. K. Palraj, ASP/IT

Innovative Practice Description

- **Unit / Topic** : Unit 4 / Routing Protocols, Unicast Protocol
- **Course Outcome** : CO4
- **Topic Learning Outcome:** TLO14
- **Activity Chosen** : Think Pair Share
- **Justification:**
 - Think-Pair-Share" (TPS) teaching strategy is particularly effective for complex topics like routing protocols and unicast protocols in computer networks due to several key factors that align with the nature of these subjects and the needs of learners.
 - This strategy is ideal for teaching Routing Protocols and Unicast Protocols because it actively engages students in critical thinking, collaboration, and problem-solving, which are essential for understanding these complex topics.
- **Time Allotted for the Activity:** 20 Minutes
- **Details of the Implementation:**
 - Instructor explained introducing a specific scenario or problem related to routing, such as determining the best path for a unicast packet in a network using a particular protocol (e.g., OSPF or RIP).
 - First, allow students 3–5 minutes to **think** individually about the factors that affect routing decisions, such as link cost, hop count, or network topology as shown in Figure 1.
 - Next, have students **pair** up and discuss their thoughts for 5–7 minutes, encouraging them to compare their reasoning, clarify doubts, and refine their understanding through peer-to-peer interaction as shown in Figure 2
 - Finally, Few students are share their findings, with each pair contributing insights on how routing decisions are made, the role of unicast addresses, or the strengths and limitations of the chosen protocol as shown in Figure 3.

• **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO4	PO5	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO2	3	3	2	2	3	1	2	2	1	2	1	1	3

(1 – Low 2 – Moderate 3 – High)

• **PO / PSO mapped:**

Innovative practice	PO1	PO2	PO3	PO4	PO5	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
	3	3	2	2	3	1	2	2	1	2	1	1	3
Justification for correlation	The students could be able to use and apply basic concept of transport layer protocol	The students could be able to review the real time problem with the transport layer protocol	The students will be able to address various problems related to transport layer	The students will be capable of understanding the various uncertainties in transport layer	The students will be capable of developing or applying various frameworks for managing transport layer	In practice, the students will adhere to ethics in various protocols in transport layer	The students will work as part of a team and individually to illustrate their competencies in the transport layer	The students will illustrate their abilities in communication and problem-solving as they engage with the transport layer	The students will be able to identify the task required to build TCP and UDP	The student will identify the technological advances made in the past and update the concepts in transport layer	The students could be able to apply computing skills to the transport layer protocols.	The students will have the ability to provide IT solutions concerning the transport layer.	The students could be able to develop software application for transport layer protocols

- Images / Screenshot of the practice



Figure 1. Pair with neighbour and discuss answer



Figure 2. S. Swathi & G. Subalakshmi share the answer to entire class

- **Reflective Critique:**

- ❖ ***Feedback of practice from students and other stakeholders:***

- Students said that the activity is very useful to apply the concept that they learnt to solve the given problem easily
- Students said that it will be very helpful when preparing for examinations.
- They felt comfortable to share their understanding about the concept with class mates

- ❖ ***Benefit of the practice:***

- Think Pair Share activity gives opportunity among students to share their understanding about the concept with neighbours and also improves their speaking and listening skills.
- This activity is very interactive and made the slow learners to clarify the doubts with their peers.
- It helps in improving students' oral communication of by delivering the concepts that they learnt in class.

- ❖ ***Challenges faced in implementation:***

- Few student pairs were randomly selected to perform their presentations in front of the class due to time constraints.
- Make the student to know the impact and significance of sharing their understanding about the topic.
- The activity needs more time to share their views

References:

- https://www.ritrjpm.ac.in/images/computer-science/2021-2022/3.VA_OCS752-ThinkPairShare.pdf
- <https://www.ritrjpm.ac.in/images/computer-science/2022-2023/TPS-RV-I-IT-PSPP.pdf>

Signature of Faculty Member

HOD